misleading and thoughtless. The Theory of Relativity, too, is based on something absolute, namely, the determination of the matrix of the space-time continuum; and it is an especially stimulating undertaking to discover the absolute which alone makes meaningful something given as relative.

Our every starting-point must necessarily be something relative. All our measurements are relative. The material that goes into our instruments varies according to its geographic source; their construction depends on the skill of the designer and toolmaker; their manipulation is contingent on the special purposes pursued by the experimenter. Our task is to find in all these factors and data, the absolute, the universally valid, the invariant, that is hidden in them,

This applies to the Theory of Relativity, too. I was attracted by the problem of deducing from its propositions that which served as their absolute immutable foundation. The way in which this was accomplished, was comparatively simple. In the first place, the Theory of Relativity confers an absolute meaning on a magnitude which in classical theory has only a relative significance: the velocity of light. The velocity of light is to the Theory of Relativity as the elementary quantum of action is to the Quantum Theory: it is its absolute core. In this connection, it turns out that a general principle of classical theory, the least-action principle, is also invariant with respect to the Theory of Relativity; accordingly, the quantum of action retains its significance in the Theory of Relativity as well.

This was what I tried to establish in all details, first for point masses, and then for black-body radiation. These researches yielded, among other results, the inertia of radiation and the invariance of entropy in systems possessing relative velocities.

But this is not all. The absolute showed itself to be even more deeply rooted in the order of natural laws than had been assumed for a long time. In 1906, W. Nernst came out with his heat theorem, often referred to as the “Third Law of Thermodynamics.” As I immediately established, it amounted to the hypothesis that entropy, until then defined only up to an additive constant, possessed an absolute positive value. This value, from which all equations of equilibrium follow, can be calculated beforehand. In the case of a chemically homogeneous solid or liquid (in other words, a solid or liquid composed of homogeneous molecules) of which the absolute temperature is zero, this value is likewise zero. This principle in itself expresses an important fact, namely that the specific heat of a solid or liquid vanishes at the absolute zero of temperature. For other temperatures, fruitful inferences follow, with respect to the melting points of a body and the transition temperature of allotropic changes. If now one passes from chemically homogeneous solids and liquids to bodies with heterogeneous molecules, or to solutions and gases, the absolute entropy is calculated by means of combinatory considerations, in